Student Attendance — Exploratory Data Analysis Report

An analysis of attendance patterns across 20,000 student records

Dataset	Attendance_Prediction.csv
Records	20,000 students · 15 features
Data quality	No missing values · No duplicate rows
Date generated	June 23, 2026

Headline finding

Class delivery format (online vs. offline) and weather conditions are the strongest observed factors associated with attendance — far outweighing demographic variables such as gender, course, or parent's education, which show almost no relationship with attendance at all.

1. Data Overview & Preparation

The dataset contains 20,000 student records with 15 columns covering demographics (age, gender, course, year), environment (parent's education, internet access, hostel residency, weather, travel time), behavior (study hours, sleep hours, class type), and the target variable, attendance (1 = present, 0 = absent).

Cleaning and feature engineering steps applied:

- Removed the **student_id** column — an identifier with no predictive value.
- Checked for missing values and duplicate rows — **none found**.
- Label-encoded categorical columns for numerical analysis.
- Engineered a **study_sleep_ratio** feature (study hours ÷ sleep hours).
- Binned travel time into a **travel_category** feature (Near / Medium / Far).
- Standardized numeric columns (age, study hours, sleep hours, travel time) for fair comparison.

2. Attendance Distribution

Attendance is well balanced across the dataset: roughly 51.6% of records are marked present and 48.4% absent. This balance is good news for any future predictive modeling — there's no skew that would require special handling.

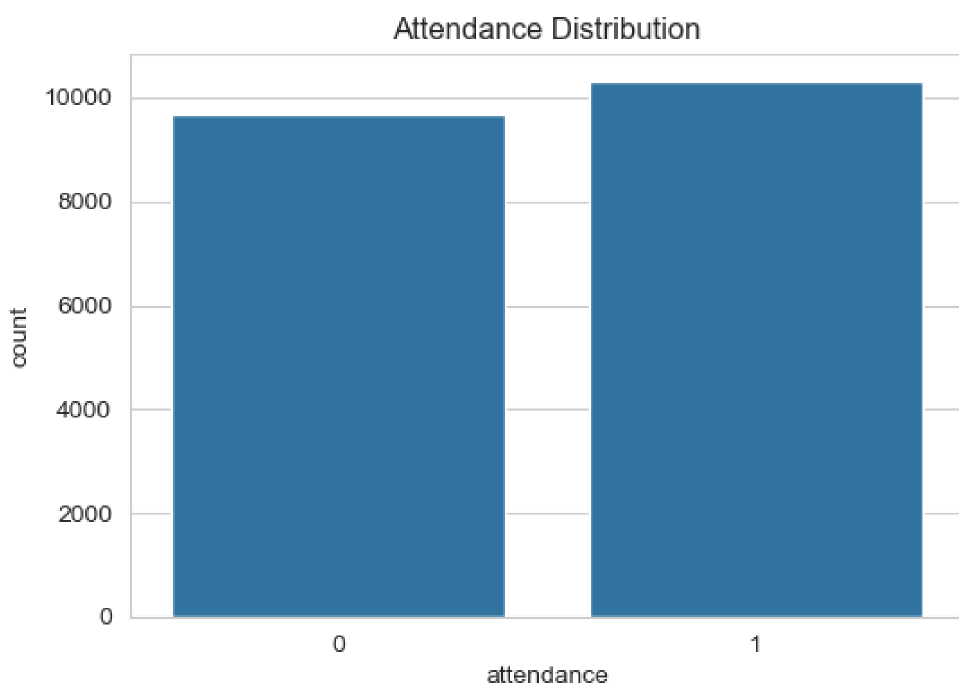

Figure 1. Overall attendance distribution (0 = Absent, 1 = Present)

3. Key Drivers of Attendance

Several factors show a meaningful relationship with attendance. They are presented below in order of impact.

3.1 Class Type — the single biggest factor

Students attending **online** classes have an attendance rate of **59.9%**, compared to **48.0%** for **offline** classes — a gap of nearly 12 percentage points. This is the largest effect found anywhere in the dataset.

3.2 Weather — rainy days hurt attendance

Attendance drops sharply on rainy days (**41.6%**) compared to sunny (54.5%), cold (54.7%), cloudy (54.0%), and hot (53.3%) conditions. Rain stands out as a clear, consistent drag on attendance.

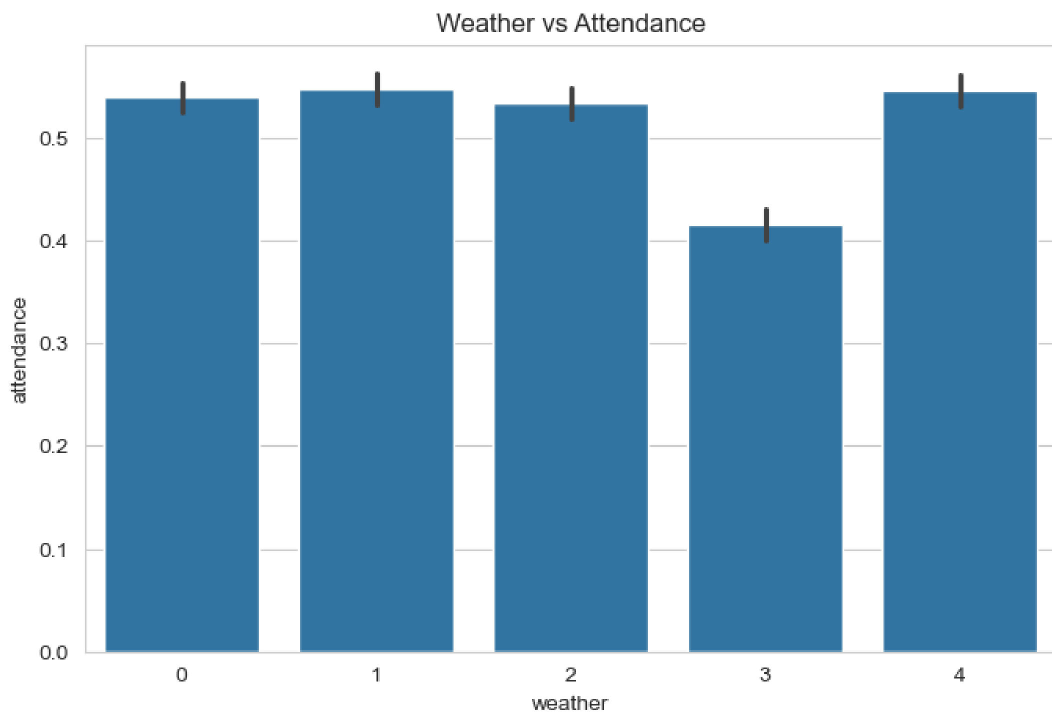

Figure 2. Average attendance rate by weather condition (x-axis codes: 0 = cloudy, 1 = cold, 2 = hot, 3 = rainy, 4 = sunny)

3.3 Study Hours — more study, more attendance

Study hours are positively correlated with attendance ($r \approx 0.20$). Students who attend class tend to report higher study hours, suggesting attendance and academic engagement reinforce each other.

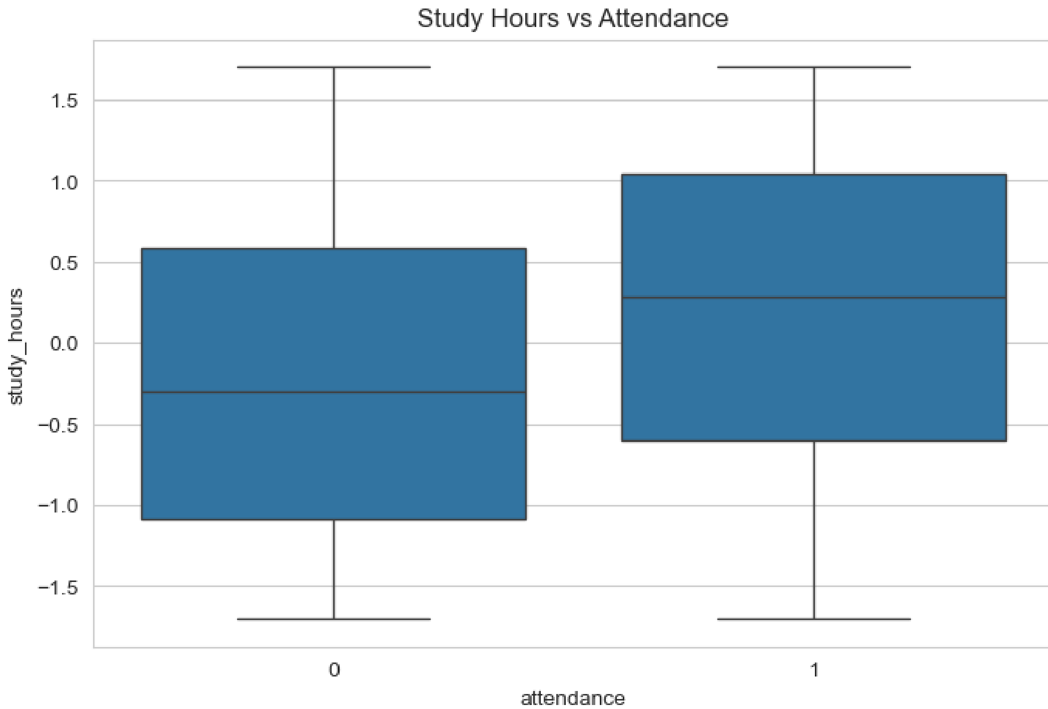

Figure 3. Distribution of study hours by attendance status

3.4 Travel Time — longer commutes, lower attendance

Travel time is negatively correlated with attendance (r ≈ -0.13). Students with longer commutes are somewhat more likely to miss class, consistent with commuting friction as a barrier to attendance.

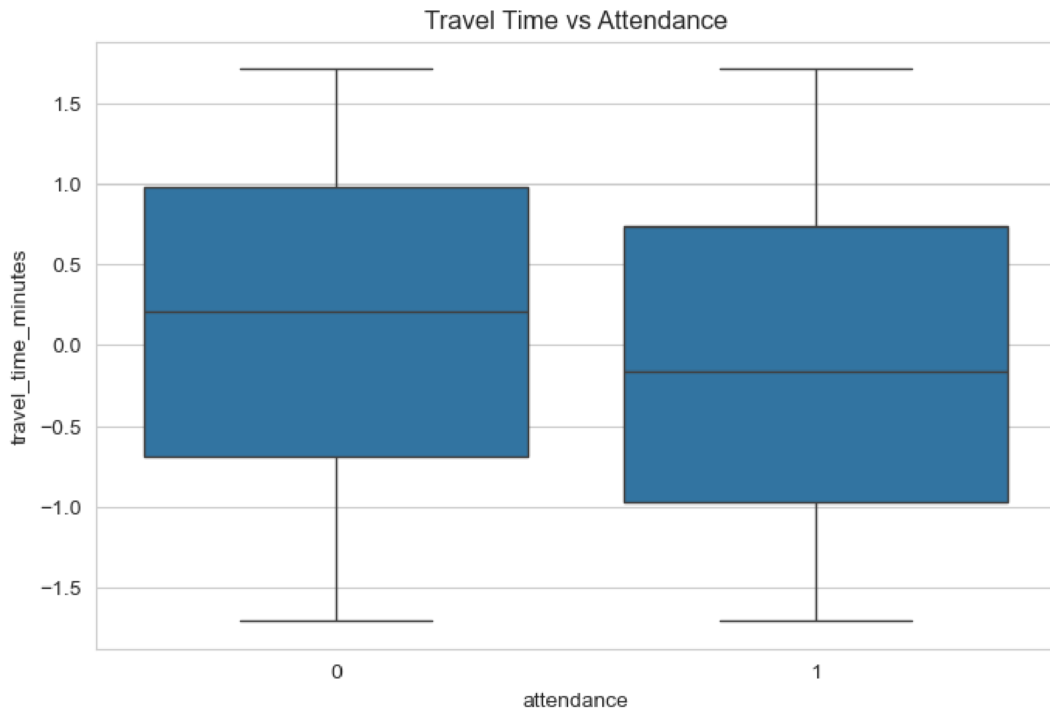

Figure 4. Distribution of commute time by attendance status

3.5 Sleep Hours — a smaller but positive effect

Sleep hours show a mild positive correlation with attendance (r ≈ 0.13) — better-rested students are slightly more likely to attend.

3.6 Demographics — minimal effect

Gender, course, year of study, parent's education, internet access, and hostel residency each show less than a 2 percentage-point spread in attendance rate between their categories. None of these demographic factors appear to meaningfully drive attendance on their own.

Factor	Range of attendance rate across categories
Gender	50.9% – 52.4%
Course	50.7% – 52.8%
Year of study	51.1% – 52.0%
Parent's education	51.0% – 53.0%
Internet access	51.5% – 52.2%
Hostel residency	51.3% – 52.2%

4. Correlation Summary

Pearson correlation of each numeric feature with attendance, ranked by strength:

Feature	Correlation with attendance
Study hours	+0.202
Sleep hours	+0.133
Age	-0.000 (no relationship)
Travel time (minutes)	-0.125

Study hours show the strongest numeric relationship with attendance, followed by travel time (negative) and sleep hours. Age shows essentially no relationship.

5. Important Data Quality Note

The absence_reason column leaks the target variable and should be excluded from any predictive model.

When absence_reason is "none," attendance is 88.5%. For every other value — sick, travel, festival, personal work, project work, no reason — attendance is exactly 0%. In other words, this column is essentially a restatement of the outcome rather than an independent predictor, since it can only be known after the fact. It should be dropped before training any attendance-prediction model, though it remains useful for understanding why absences happen.

6. Conclusions & Recommendations

- **Prioritize class format and weather contingencies.** The 12-point attendance gap between online and offline classes, and the dip on rainy days, suggest that flexible or remote options (e.g., recorded lectures, hybrid sessions) on poor-weather days could meaningfully improve attendance.
- **Support students with long commutes.** The negative correlation between travel time and attendance points to commuting as a real barrier — flexible arrival policies or transport support may help.
- **Demographic targeting is not supported by this data.** Gender, course, year, parent's education, internet access, and hostel residency show negligible differences in attendance — targeted interventions by these groups are unlikely to move the needle.
- **Exclude absence_reason from predictive models.** Use it only for diagnostic or root-cause analysis of absences, not as a model input.
- **Study and sleep habits track with attendance.** Whether as cause or effect, engagement (study hours) and rest (sleep hours) move together with attendance and could be worth monitoring as early indicators.